

Exp. No. 6

Design a Simple Traffic Light Controller

Aim

To design and simulate a simple Traffic Light Controller using Verilog HDL in the Xilinx Vivado environment.

Verilog Code

```
module traffic_light (
    input clk,
    input reset,
    output reg red,
    output reg yellow,
    output reg green
);

reg [1:0] state;
reg [3:0] timer;

parameter RED    = 2'b00,
          GREEN   = 2'b01,
          YELLOW  = 2'b10;

always @ (posedge clk or posedge reset) begin

    if(reset) begin
        state <= RED;
        timer <= 0;
    end

    else begin
        timer <= timer + 1;
        case(state)
            RED:
                if(timer == 5) begin
                    state <= GREEN;
                    timer <= 0;
                end
            GREEN:
                if(timer == 5) begin
                    state <= YELLOW;
                    timer <= 0;
                end
            YELLOW:
                if(timer == 2) begin
                    state <= RED;
                    timer <= 0;
                end
        endcase
    end
end

always @(*) begin
    red = 0;
    yellow = 0;
    green = 0;
    case(state)
        RED:    red = 1;
        GREEN:  green = 1;
        YELLOW: yellow = 1;
    endcase
end
endmodule
```

Testbench Code

```
module tb;

reg clk;
reg reset;
wire red;
wire yellow;
wire green;

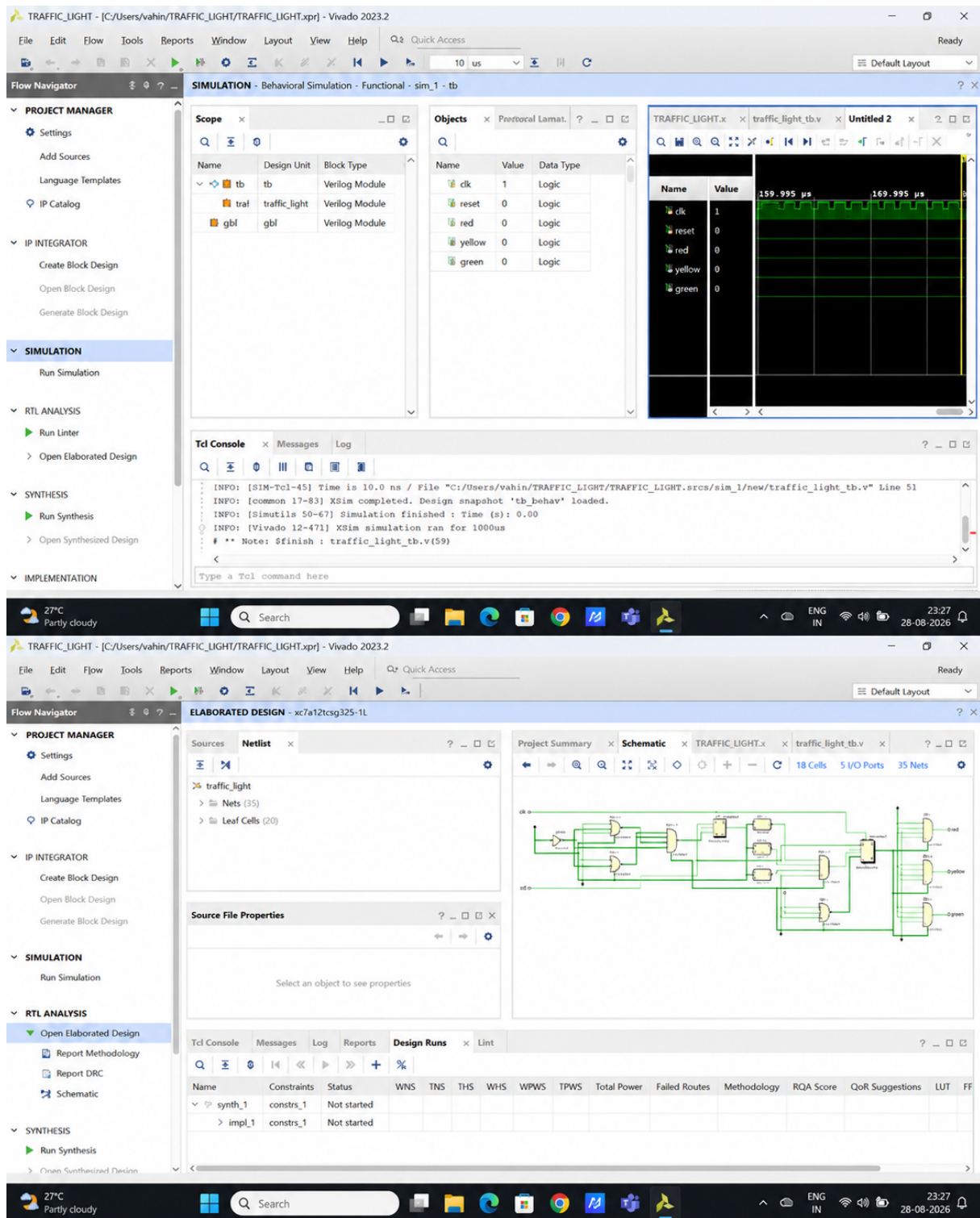
traffic_light uut (
    .clk(clk),
    .reset(reset),
    .red(red),
    .yellow(yellow),
    .green(green)
);

always #5 clk = ~clk;

initial begin
    clk = 0;
    reset = 1;
    #10 reset = 0;
    #100;
    $finish;
end

endmodule
```

Output (Simulation Waveform / Schematic)



Result: The Traffic Light Controller was designed and simulated successfully. On reset, the state machine initializes to RED, and cycles through RED → GREEN → YELLOW → RED based on the internal timer, driving the red, yellow, and green outputs one at a time as verified in the simulation waveform and elaborated schematic.